Deg4 partitions

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Lemma 0.1. *Given a partition μ, a strict β-expansion of γ, Then the greatest such γ is when γ is μ and $\beta = \varnothing$*

Proof. Feels obvious, but strict expansion means that $N^{\mu}_{\gamma,\beta} \neq 0$. But $N^{\mu}_{\mu,\beta} \neq 0 \iff \beta = \varnothing$. The degree of μ should equal the sum of both. That is $|\mu| = |\gamma| + |\beta|$ Then γ is at most $|\mu|$ and $|\beta| = 0 \implies \beta = \varnothing$, □

Proposition 0.2. *If $\mu \subseteq \lambda$ is a subpartition of λ (contained as a diagram), then the last non zero socle layer is at:*

$$\mathcal{K} = |\lambda \cap \mu| + |\gamma_2|$$

and γ_2 is a maximal sub diagram of λ and μ, and λ is a strict γ_2-expansion of $\lambda \cap \mu$

Proof. We search for the maximum k. We already know that for λ and μ, $K_{max} \leq \min\{\lambda, \mu\} = |\mu| = |\lambda \cap \mu|$ Then of course

$$K_{max} \leq |\lambda \cap \mu| \leq |\lambda \cap \mu| + |\gamma_2|$$

$$K_{max} \leq |\lambda \cap \mu| + |\gamma_2|$$

For any γ_2 Partition. So true for a γ_2 like the one mentioned above.

For the other direction, recall that for any $k = |\gamma_1| + |\gamma_2|$ non-zero, there exists partitions α, β such that λ, μ are strict expansions of δ_1 respectively. Equivalently, this means that $N^{\lambda}_{\gamma_1,\alpha} \cdot N^{\lambda}_{\gamma_2,\alpha} \neq 0$. Since μ is a subdiagram of λ then any such $\beta \subseteq \mu$ is also a subdiagram of λ.Using Lemma 0.1 we can infer that a maximal expansion to reach μ is when $\gamma = \mu = \lambda \cap \mu$ and consequently $\beta = \varnothing$.

$$\implies |\lambda \cap \mu| = |\gamma_1| \leq |\gamma_1| + |\gamma_2| \quad \text{For any } \gamma_2 \text{ subdiagram of } \lambda \cap \mu.$$

$$\implies |\gamma_1| + |\gamma_2| = |\lambda \cap \mu| + |\gamma_2| \leq \max_{\gamma_2} |\gamma_1| + |\gamma_2| = K_{max}$$

$$\iff |\lambda \cap \mu| + |\gamma_2| \leq K_{max}$$

Then both inequalities guarantee $|\lambda \cap \mu| + |\gamma_2| = K_{max}$. □

Definition 0.3. *Let γ_1, γ_2 be subpartitions of both λ and μ. Then we say that (γ_1, γ_2) is subordinate to (λ, μ) if both λ and μ contain some strict γ_2-expansion of γ_1.*

Proposition 0.4. *For partitions λ, μ, the maximum non-zero socle layer is*

$$K = \max_{\gamma_1,\gamma_2} |\gamma_1| + |\gamma_2|$$

over all (γ_1, γ_2) subordinate pairs to (λ, μ).

Proof. Let k be the last non-zero socle layer that is: $\overline{\text{soc}}^k I^{\lambda,\mu} \neq 0$. That is, there exists λ', μ' subpartitions of λ, μ respectively such that $N^{\lambda}_{\gamma_1,\gamma_2,\lambda'} \cdot N^{\lambda}_{\gamma_1,\gamma_2,\mu'} \neq 0$ For some γ_1, γ_2 subdiagrams of both λ, μ. Equivalently, $N^{\lambda}_{\gamma_1,\gamma_2,\lambda'} \neq 0$ and $N^{\lambda}_{\gamma_1,\gamma_2,\mu'} \neq 0$. These coefficients are

$$N^{\lambda}_{\gamma_1,\gamma_2,\lambda'} = \sum_{\nu} N^{\nu}_{\gamma_1,\gamma_2} N^{\lambda}_{\nu,\lambda'}$$

$$N^{\mu}_{\gamma_1,\gamma_2,\mu'} = \sum_{\nu'} N^{\nu'}_{\gamma_1,\gamma_2} N^{\mu}_{\nu',\mu'}.$$

Note that since both are non-zero the coefficients $N^{\nu'}_{\gamma_1,\gamma_2}$ and $N^{\nu}_{\gamma_1,\gamma_2}$ are non-zero for some ν, ν'. This is equivalent as saying that both ν, ν' are a strict γ_2-expansion of γ_1. and additionally $\nu \subseteq \lambda$ (since $N^{\lambda}_{\nu,\lambda'} \neq 0$) and similarly $\nu' \subseteq \mu$. Then (γ_1, γ_2) are subordinate to (λ, μ). □

0.1 The One-Row Case

Theorem 0.5 (One-row formula)**.** *Let $\lambda = (a)$, where $a \geq 0$, and let μ be an arbitrary partition. Then*

$$\boxed{k_{\max}((a), \mu) = \min\{a, \mu_1 + \mu_2\}.} \tag{1}$$

Proof. Suppose that k is a nonzero layer. Then there exist partitions $\nu \subseteq (a)$ and $\omega \subseteq \mu$, both of size k, and partitions δ, γ such that

$$N^{\nu}_{\delta,\gamma} > 0, \qquad N^{\omega}_{\delta,\gamma} > 0.$$

Since ν is a single-row partition, the containment properties of Littlewood–Richardson coefficients imply that δ and γ must also be single-row partitions. Write

$$\delta = (d), \qquad \gamma = (g), \qquad d + g = k.$$

By Pieri's rule,

$$s_{(d)} s_{(g)} = \sum_{j=0}^{\min(d,g)} s_{(d+g-j,j)}.$$

Thus every constituent of $s_\delta s_\gamma$ has at most two rows. In particular, $\ell(\omega) \leq 2$, and since $\omega \subseteq \mu$,

$$k = |\omega| \leq \mu_1 + \mu_2.$$

Moreover, $\nu \subseteq (a)$ implies $k \leq a$. Consequently,

$$k_{\max}((a), \mu) \leq \min\{a, \mu_1 + \mu_2\}.$$

To establish the reverse inequality, set

$$k = \min\{a, \mu_1 + \mu_2\}.$$

Choose a two-row subpartition $\omega = (u, v) \subseteq \mu$ with $u + v = k$. Such a partition exists by removing boxes from the first two rows of μ.

Take

$$\delta = (u), \qquad \gamma = (v).$$

Pieri's rule shows that both (k) and (u, v) occur in the product $s_\delta s_\gamma$. Hence

$$N_{\delta,\gamma}^{(k)} > 0, \qquad N_{\delta,\gamma}^{\omega} > 0.$$

Since $(k) \subseteq (a)$ and $\omega \subseteq \mu$, the degree k is attained.

Therefore,

$$k_{\max}((a), \mu) = \min\{a, \mu_1 + \mu_2\}.$$

□

Corollary 0.6 (One-column formula)**.** *For $\lambda = (1^a)$ and arbitrary μ,*

$$\boxed{k_{\max}((1^a), \mu) = \min\{a, \mu_1^{\mathsf{T}} + \mu_2^{\mathsf{T}}\}.}$$

Proof. This follows from the one-row theorem and the conjugation identity

$$N_{\delta,\gamma}^{\nu} = N_{\delta^{\mathsf{T}},\gamma^{\mathsf{T}}}^{\nu^{\mathsf{T}}},$$

which implies

$$k_{\max}(\lambda, \mu) = k_{\max}(\lambda^{\mathsf{T}}, \mu^{\mathsf{T}}).$$

□